IT2011
AI and ML

Lecture 01

Introduction to Artificial Intelligence and Machine Learning

Year 02
Semester 01

"A Better Future with AI!" ☺

1 What is Artificial Intelligence (AI)?

- AI is the field of computer science that focuses on creating machines or systems that can perform tasks that normally require human intelligence.
- These tasks include:
  - Learning
  - Reasoning
  - Problem solving
  - Understanding natural language
  - Making decisions

"Make machines think and act like humans!"

Examples of AI in real life



Virtual assistants (Siri, Alexa)



Self-driving cars



Recommendation systems (Netflix, YouTube)

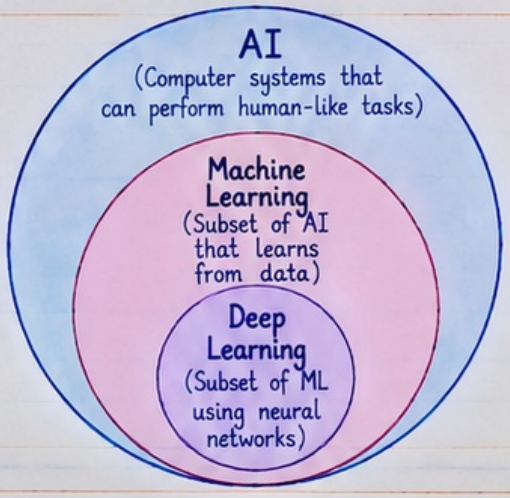


Medical diagnosis systems

2 What is Machine Learning (ML)?

- Machine Learning is a subset of AI.
- ML focuses on giving computers the ability to learn from data without being explicitly programmed.
- Instead of writing step-by-step instructions, we provide data and let the system find patterns and make predictions or decisions.

"Learn from data, not from instructions!"



3 Traditional Programming vs AI/ML

Traditional Programming

Input (data) → Rules (Program) → Output (result)

- We give the rules and logic explicitly.
- Computer follows the rules to produce output.
- Example: A calculator program.

Machine Learning

Input (data) → Learning (Algorithm) → Output (prediction/ (decision)

- We give data and expected output.
- The system learns the rules (patterns) from data.
- Example: Spam email detection, image recognition.

4 Key Differences: AI vs ML

Feature	Artificial Intelligence (AI)	Machine Learning (ML)
Definition	Machines that can perform human-like tasks	Subset of AI that learns from data
Scope	Broader	Narrower (part of AI)
Approach	Can use rules, logic, search, etc.	Uses data and algorithms
Learning	Not always required	Always learns from data
Examples	Robotics, expert systems, chatbots	Image recognition, recommendation systems

5 Why is AI/ML Important?

- Automates repetitive tasks
- Solves complex problems
- Helps in decision making
- Finds patterns in large amounts of data
- Used in many real-world applications
- Improves efficiency and accuracy

"AI and ML are transforming the way we live, work and learn!"

6 Final Takeaway

AI is the big picture. • ML is a powerful tool inside AI.

Both are shaping the future and create many opportunities! ☺

"Remember!

- ★ AI = Human-like intelligence
- ★ ML = Learn from data
- ★ ML is a subset of AI
- ★ Deep Learning is a subset of ML

Lecture 01 - Step 2

Artificial Intelligence (AI) vs Machine Learning (ML)

t is Artificial Intelligence (AI)?

a broad field of computer science
ocuses on creating machines that
rform tasks that normally require
intelligence.

tasks include reasoning, problem
, planning, understanding natural
ge, perception, and decision making.

s: expert systems, robotics,
language processing, game playing

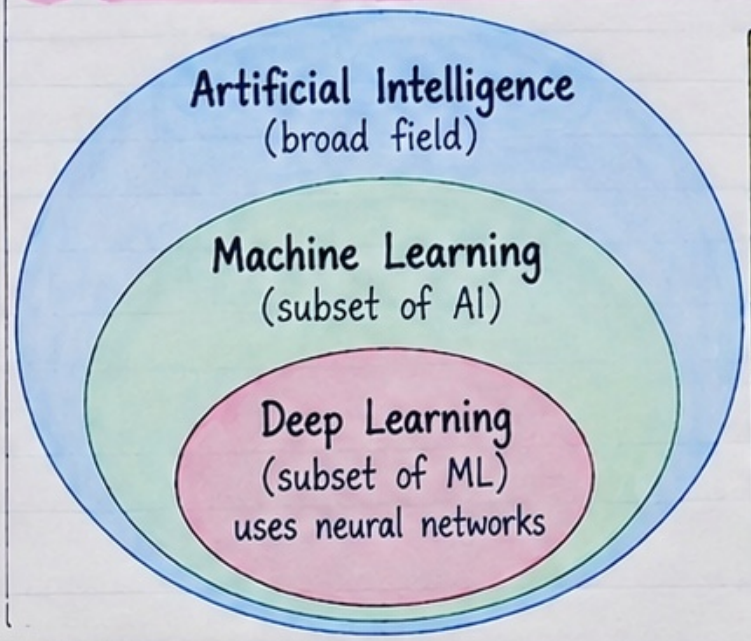
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a subset of AI that allows computers
n patterns from data without being
tly programmed.

l of writing rules, we give data
les) and the system learns a model
ke predictions or decisions.

s: email spam detection, image
ion, recommendation systems

③ Relationship between AI, ML and DL



All M
Learn
but
Mach

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is M
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④ Key Differences

Aspect	Artificial Intelligence (AI)	Machine Le
Scope	Broad field	Subset of
Approach	Can use rules, logic, search, planning, ML etc.	Learns fro
How it works	Can be rule-based or data-driven	Data-drive patterns)
Goal	Perform human-like tasks	Improve pe from exper
Examples	Expert systems, robots, game playing, NLP	Regression, clustering,

is ML important?

ndle complex problems.
from real-world data.
automatically with more data.
many real applications today.



Exam Key Points

- AI = broad field, human-like intelligence.
- ML = subset of AI, learns from data.
- Deep Learning = subset of ML, uses neural
- All ML is AI, but not all AI is ML.
- Give one or two examples for each.

aims to make machines smart. ML gives machines the ability to learn

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Lecture 01 (Part 3)

Applications, Benefits, Limitations, Ethics and Future of AI/ML

Year 02
Semester 01

"Learn Today
Lead Tomorrow !"

1 Real-world Applications of AI/ML

- AI and ML are used in many areas in our daily life.

Area	Examples
Healthcare	Disease detection, medical imaging, personalized treatment
Transportation	Self-driving cars, traffic prediction
Business	Recommendation systems, customer analysis
Communication	Chatbots, language translation
Security	Face recognition, fraud detection
Agriculture	Crop monitoring, yield prediction
Entertainment	Game AI, content recommendation
Manufacturing	Quality control, predictive maintenance

AI is already around us in many ways !

2 Benefits of AI/ML

- Can automate repetitive and time-consuming tasks
- Can analyze large amounts of data quickly
- Helps to make better and faster decisions
- Can find patterns that humans may miss
- Improves efficiency and reduces cost
- Works 24/7 without getting tired
- Can perform dangerous tasks instead of humans

" AI / ML makes our work easier, faster and smarter ! "

3 Limitations and Challenges

- Needs large amounts of data
- Quality of data is very important
- Can be biased if the data is biased
- May not always give correct results
- Can be expensive and require high resources (computing power, storage)
- Some tasks are still difficult for AI (e.g. common sense, real understanding)
- Lack of transparency (hard to explain decisions)

AI is powerful, but it is not perfect !

4 AI Ethics and Responsible Use

- Avoid bias and treat everyone fairly
- Protect privacy and personal data
- Be transparent about how AI systems work
- Use AI for good and not for harmful purposes
- Consider the impact on jobs and society
- Follow laws and regulations
- As future professionals, we should use AI responsibly

Use AI for a better and fairer world !

" With great power comes great responsibility ! "

5 Future of AI/ML

- More advanced and human-like systems
- More use in healthcare, education, agriculture, etc.
- Better and safer AI models
  Growth of generative AI (text, image, video, etc.)
- More job opportunities in AI-related fields
- Need for responsible development and regulations

AI will continue to shape a smarter and better future !

6 Summary of Lecture 01

- AI is a field of computer science to create intelligent machines.
- ML is a subset of AI that learns from data.
- There are different branches of AI and types of ML.
- Deep Learning uses neural networks with many layers.
- Generative AI can create new content.
- Prompt Engineering helps us get better results from AI tools.
- AI/ML have many real-world applications.
- They bring many benefits but also have limitations and challenges
- It is important to use AI responsibly and ethically.
- AI will continue to grow and create new opportunities in the future.

Understand
Use
Be Responsible
Be Part of a
Better Future !

Key Takeaways

1. AI = Human-like intelligence
2. ML = Learn from data
3. DL = Subset of ML
4. Generative AI = Create new content
5. Prompt Engineering = Better results
6. AI has benefits and challenges
7. Responsible use is important
8. AI will shape the future

Small Steps Big Future !

" Artificial Intelligence is not just a technology, it is a powerful tool to create a better tomorrow ! "

Keep Learning !
Keep Growing !

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Lecture 01 (Part 4)

Summary, Exam Focus and Key Points

Year 02
Semester 01

"Study Today
Succeed Tomorrow !"

1 Key Definitions (Quick Revision)

Term	Definition
Artificial Intelligence (AI)	Field of computer science that focuses on creating machines that can perform human-like tasks.
Machine Learning (ML)	Subset of AI that learns from data without being explicitly programmed.
Deep Learning (DL)	Subset of ML that uses artificial neural networks with many layers.
Generative AI	Type of AI that can create new content (text, images, audio, video).
Prompt Engineering	Process of giving the right instructions (prompts) to AI models to get better results.

Key Definitions are very important for exams !

2 AI/ML Learning Journey

Data (Collect data from real world)
↓
Preprocessing (Clean and prepare data)
↓
Model / Algorithm (Choose and train a model)
↓
Evaluation (Check performance)
↓
Prediction / Decision (Use the model in real world)

A simple view of how AI/ML works !

3 Real-world Example (Complete View)

Example: Email Spam Detection

Emails (Input) → Preprocessing (Remove noise, convert to numbers, etc.) → ML Algorithm (e.g. Naive Bayes, Logistic Regression) → Model (Learn from labelled data) → Output: Spam or Not Spam

This process is similar for many real-world AI/ML applications !"

4 Common Exam Questions

1. Define AI, ML, Deep Learning, Generative AI and Prompt Engineering.
2. Differentiate between AI and ML.
3. List and explain the branches of AI.
4. Explain the types of Machine Learning.
5. Give real-world examples for AI/ML applications.
6. Discuss the benefits and limitations of AI/ML.
7. Explain why prompt engineering is important.
8. Draw and explain a simple AI/ML workflow.
9. Discuss the ethical issues of AI.
10. Write a short note on the future of AI/ML.

Practice these types of questions !

5 Advantages and Disadvantages

Advantages	Disadvantages
✓ Can automate repetitive tasks	✗ Needs large amounts of data
✓ Can analyze large amounts of data	✗ Can be expensive (hardware, data)
✓ Helps in better and faster decisions	✗ May produce biased results
✓ Improves accuracy and efficiency	✗ Lack of transparency (hard to explain)
✓ Useful in many real-world applications	✗ Can be misused (e.g. deepfakes)
✓ Can solve complex problems	✗ May replace some human jobs
	✗ Still cannot fully match human intelligence

6 Tips for Studying Lecture 01

1. Understand the basic definitions clearly.
2. Draw and remember the diagrams.
3. Learn real-world examples.
4. Focus on the differences (AI vs ML, etc.).
5. Be ready to write short notes.
6. Practice past paper questions.
7. Use your own words to explain.
8. Make short summaries for quick revision.

Understand Don't just memorize !

7 Final Summary

- AI is about making machines think and act like humans.
- ML is a subset of AI that learns from data.
- There are different branches and types of AI/ML.
- Deep Learning uses neural networks with many layers.
- Generative AI creates new content.
- Prompt Engineering helps us get better results from AI tools.
- AI/ML have many real-world applications, benefits and challenges.
- It is important to use AI responsibly and ethically.
- AI/ML will continue to grow and shape the future.

A strong foundation today leads to a brighter tomorrow !

Keep in Mind !

- AI is a tool, not a replacement for humans.
- Good data leads to good results.
- Use AI for good and positive purposes.
- Always think about the ethical impact.
- The future of AI depends on how we use it.
- Keep learning and stay curious !

" Learn Today → Build Skills → Create a Better Tomorrow ! "

Good Luck !

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Lecture 01 (Part 5)

Extra Notes, Revision Map and Mind Map

Year 02
Semester 01

"Small Steps
Lead to Big Knowledge !"

1 Key Terminologies (Extra)

Term	Meaning (Simple)
Data	Raw facts and figures (numbers, text, images, etc.)
Information	Processed data that is meaningful.
Model	A learned pattern from data.
Algorithm	A set of steps to solve a problem.
Training	Process of learning patterns from data.
Testing	Process of checking how well the model works on new data.
Prediction	Output given by the model for new data.
Feature	An input variable (e.g, age, price, size).
Label	The correct output (answer) used in training.
Dataset	A collection of data examples.

"Clear terms make everything easier !"

2 AI/ML in Daily Life

- Search engines (Google) → AI/ML ranks results
- Email spam filter → ML classifies spam or not
- Maps and navigation (Google Maps) → ML predicts traffic
- Video recommendation (YouTube) → ML suggests videos
- Online shopping (Amazon) → Recommendation systems
- Self-driving cars → AI + ML + Computer Vision
- Language translation (Google Translate) → NLP
- Voice assistants (Siri, Alexa) → Speech recognition
- Healthcare → Disease prediction and medical imaging

"AI/ML is not just theory, it is all around us !"

3 AI vs Human Intelligence

Aspect	AI/ML	Human Intelligence
Learning	Learns from data	Learns from experience
Speed	Very fast	Slower
Accuracy	High (for specific tasks)	Can make mistakes
Emotions	No emotions	Has emotions
Creativity	Limited (but improving)	Highly creative
Common Sense	Limited	Strong
Adaptability	Good for specific tasks	Can adapt to many situations
Examples	Image recognition, spam detection	Understanding jokes, complex reasoning

AI is powerful, but it does not replace humans.
It works best when we use it together with human intelligence !

4 Common Misconceptions

Misconception	Reality
✗ AI is the same as ML.	✓ ML is a subset of AI.
✗ AI will replace all human jobs.	✓ AI helps humans and creates new opportunities.
✗ AI can think like a human.	✓ AI can perform tasks, but does not truly think or have emotions.
✗ AI is always 100% accurate.	✓ AI can make mistakes.
✗ AI is only used in big companies.	✓ AI is used in many small and large organizations.
✗ AI is only about robots.	✓ AI includes many areas like NLP, vision, recommendation, etc.

"Understanding the truth helps us use AI/ML wisely !"

5 Steps to Build an AI/ML Project

1. **Problem Definition**
   - Understand the problem
   - Set the goal
2. **Data Collection**
   - Collect data from real world
3. **Data Preprocessing**
   - Clean data
   - Handle missing values
   - Convert to suitable format
4. **Model Training**
   - Choose algorithm
   - Train the model using data
5. **Evaluation**
   - Check accuracy
   - Use metrics (e.g. accuracy, precision)
6. **Prediction / Use**
   - Use the model for real-world predictions

"Follow these steps for any AI/ML problem !"

6 Final Exam Focus (Lecture 01)

- [x] Definitions: AI, ML, DL, Generative AI, Prompt Engineering
- [x] Branches of AI and types of ML
- [x] Differences between AI and ML
- [x] Real-world applications and examples
- [x] Basic working process of AI/ML
- [x] Advantages and disadvantages
- [x] Ethical issues and limitations
- [x] Future of AI/ML
- [x] Short answer and explanation questions
- [x] Be able to draw simple diagrams

"Revise Smart Score High !"

Keep Learning
Keep Growing !

"Knowledge of AI today creates more opportunities tomorrow !"

All the Best !

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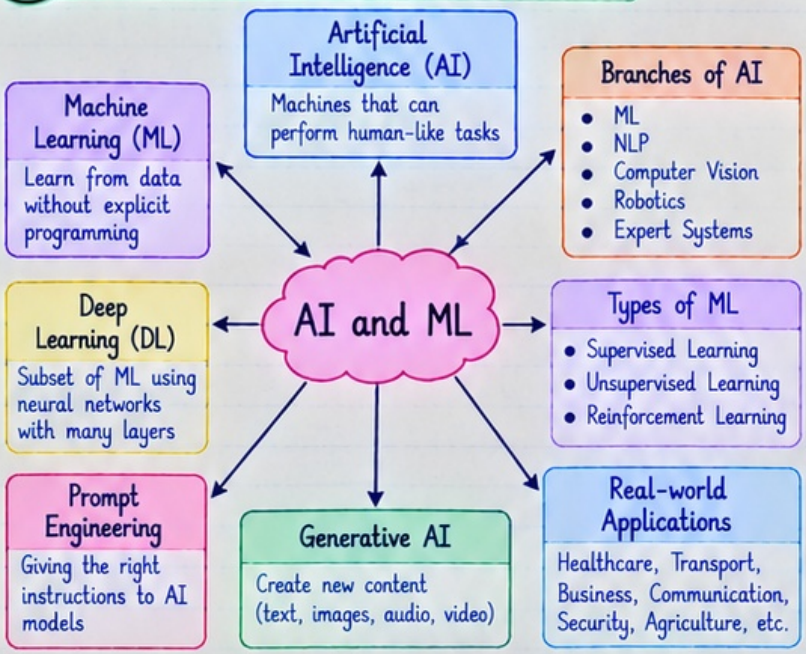
Lecture 01 (Part 6)

Mind Map, Quick Revision and Important Points

Year 02
Semester 01

"Revise Today
Be Ready Tomorrow !" ☺

1 Complete Mind Map (Lecture 01)



All these topics are connected and help us understand how AI/ML works in the real world !" ☺

2 Quick Revision Summary

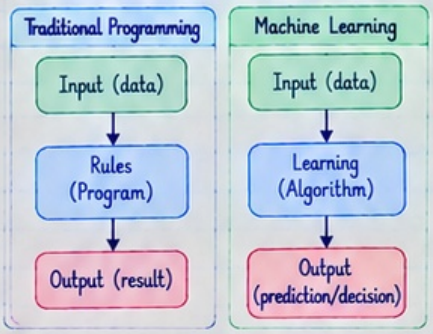
1. AI is the field of computer science that focuses on creating machines that can perform human-like tasks.
2. ML is a subset of AI that learns from data.
3. DL is a subset of ML that uses neural networks with many layers.
4. Generative AI can create new content (text, images, audio, video).
5. Prompt engineering helps us get better results from AI models.
6. AI has many real-world applications in our daily life.
7. AI can automate tasks, analyze large data, and help in decision making.
8. AI also has limitations, ethical issues and challenges.
9. It is important to use AI responsibly and ethically.
10. AI/ML will continue to grow and create many opportunities in the future.

Small Revision Big Results ! ☺

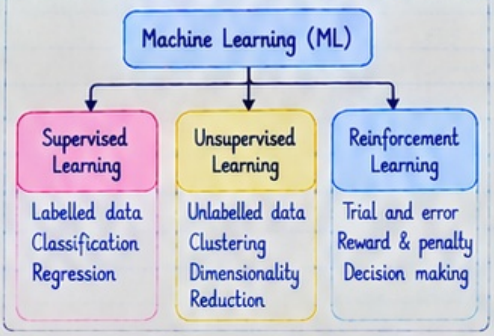
"Artificial Intelligence is not just a technology, it is a powerful tool to create a better tomorrow !" ☺

3 Important Diagrams to Remember

Traditional Programming vs AI/ML



Types of Machine Learning



4 Key Formulas / Concepts

- No complex formulas in Lecture 01.
- Focus on understanding definitions, concepts and examples.
- Be able to explain differences (AI vs ML, etc.).
- Be able to draw simple diagrams.
- Remember real-world applications.
- Understand advantages, disadvantages and limitations.
- Know the ethical issues of AI.
- Be ready to answer short and explanation questions.

Focus on Concepts not Formulas (in this lecture ! ☺

5 Possible Exam Questions (Examples)

1. What is Artificial Intelligence? Give examples.
2. What is Machine Learning? How is it different from AI?
3. Explain the branches of AI.
4. What are the types of Machine Learning? Give examples.
5. What is Deep Learning? How does it work?
6. What is Generative AI? Give some real-world examples.
7. What is prompt engineering? Why is it important?
8. List some real-world applications of AI/ML.
9. What are the advantages and disadvantages of AI/ML?
10. Discuss the ethical issues and limitations of AI.

Practice Past Papers and Try to Write in Your Own Words ! ☺

6 Study Tips for Success

- ☑ Read and understand each concept clearly.
- ☑ Look at the diagrams and try to redraw them.
- ☑ Make short notes and summaries.
- ☑ Try to explain the concepts in your own words.
- ☑ Practice past paper questions.
- ☑ Discuss with friends and solve questions together.
- ☑ Focus more on understanding, not memorizing.
- ☑ Use real-world examples to remember concepts.
- ☑ Keep revising regularly.
- ☑ Be confident and do your best !

You Can Do It ! ☺

7 Final Takeaway

- AI and ML are transforming the world.
- They bring many benefits and opportunities.
- They also have challenges and responsibilities.
- As future professionals, we must use AI wisely and ethically.
- Keep learning and stay curious !

Learn Apply Grow Succeed ! ☺

"Small steps in learning today lead to big opportunities tomorrow !" ☺

End of Lecture 01
Keep Going !

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Lecture 01 (Part 7)

Final Revision, Practice Questions and Exam Tips

Year 02
Semester 01

"Revision Today
Success Tomorrow !" ☺

1 Complete Summary of Lecture 01

1. Introduction to AI and ML
2. Traditional Programming vs AI/ML
3. AI vs ML
4. Branches of AI (ML, NLP, CV, Robotics, Expert Systems)
5. Types of ML (Supervised, Unsupervised, Reinforcement)
6. Deep Learning (Neural Networks)
7. Generative AI (text, images, audio, video)
8. Prompt Engineering
9. Applications, benefits, limitations, ethics
10. Future of AI/ML

All topics are connected and help us understand how AI/ML works in the real world ! ☺

2 Comparison Table (Quick Revision)

Aspect	Traditional Programming	AI / Machine Learning
Input	Data + Rules (Program)	Data (Examples)
How it works	Explicitly programmed by humans	Learns patterns from data
Output	Exact result (as per rules)	Prediction / Decision
Handles new data	Only if new rules are added	Can generalize from learned data
Flexibility	Less flexible	More flexible
Best for	Well-defined problems	Complex, real-world problems
Example	Calculator, payroll system	Image recognition, spam detection

" Traditional programming tells the computer what to do,
AI/ML helps the computer learn what to do ! " ☺

3 Real-world Examples (Revision)

Area	AI/ML Application	Type	Example
Healthcare	Disease prediction	ML (Supervised)	Predict diabetes
Transportation	Self-driving cars	ML + AI	Detect objects
Business	Recommendation	ML (Unsupervised)	Suggest products
Communication	Language translation	NLP	Google Translate
Security	Fraud detection	ML (Supervised)	Detect fake transactions
Agriculture	Crop monitoring	Computer Vision	Predict crop yield
Entertainment	Game AI	Reinforcement Learning	Game playing agents

"AI/ML is already part of our daily life, and it will be even bigger in the future !" ☺

4 Advantages, Disadvantages and Challenges

Advantages

- ✓ Can automate repetitive tasks
- ✓ Can analyze large amounts of data
- ✓ Helps in better and faster decisions
- ✓ Useful in many real-world applications
- ✓ Can solve complex problems
- ✓ Improves efficiency and reduces cost

Disadvantages / Challenges

- ✗ Needs large amounts of data
- ✗ Can be expensive (hardware, data)
- ✗ May produce biased results
- ✗ Can be misused (deepfakes, fraud)
- ✗ Lack of transparency (hard to explain)
- ✗ Can still make mistakes
- ✗ Raises ethical and privacy concerns

Key Point: → AI/ML are powerful tools, but we must use them wisely and responsibly ! ☺

5 Short Answer Practice Questions

1. What is Artificial Intelligence ? Give an example.
2. What is Machine Learning? How is it different from AI?
3. Explain the branches of AI with examples.
4. What are the types of Machine Learning?
5. What is Deep Learning? Why is it useful?
6. What is Generative AI? Give some examples.
7. What is Prompt Engineering? Why is it important ?
8. List some real-world applications of AI/ML.
9. What are the advantages and disadvantages of AI/ML?
10. Discuss the ethical issues of AI. How can we use AI responsibly ?

Try to answer these questions in your own words ! ☺

6 Study Plan and Exam Tips

- ☑ Go through all the notes and diagrams.
- ☑ Understand the basic definitions clearly.
- ☑ Draw the important diagrams from memory.
- ☑ Practice past paper questions.
- ☑ Try to explain the concepts in your own words.
- ☑ Focus on real-world examples.
- ☑ Make short summary notes.
- ☑ Discuss with friends and clear your doubts.
- ☑ Do not just memorize – understand !
- ☑ Be confident and stay calm in the exam !

Plan
Practice
Be Confident
You Can
Do It ! ☺

Final Words

- AI and ML are future skills – keep learning !
- Use AI tools for good and positive purposes.
- Always think about the impact on people and society.
- This is just the beginning – there is much more to explore !
- Believe in yourself and keep growing !

Learn
Today
Build a Better
Tomorrow ! ☺

You Did It !

This is the end of Lecture 01 notes.
Keep revising, keep practicing and keep learning !
All the best for your exam ! ☺

" A small step today can lead to a big future tomorrow ! " ☺

Good Luck !